

ORF 524 - Fall 2024

Week 05

$$X_n \rightarrow_P X \quad \text{i p f} \quad \forall \varepsilon > 0$$

$$\lim_{n \rightarrow \infty} P \left[\underbrace{d(X_n, X)}_{\text{r.v.}} > \varepsilon \right] = 0$$

$$d(a, b) = \|a - b\|$$

Theorem:

- Markov:

$$P[\|A\| > t] \leq \frac{E[\|A\|]}{t}$$

- Sub-Exponential
Inequality

$$P[\|A\| > t] \leq \exp\left(-\frac{c}{t}\right)$$

WLLN

$Y_1, Y_2, \dots, Y_n$ iid

$E[|Y_i|^2] < \infty$

$$\bar{Y}_n \rightarrow_P E[Y_i] = \mu$$

(as $n \rightarrow \infty$)

Proof:

$$\lim_{n \rightarrow \infty} P[|\bar{Y}_n - \mu| > \varepsilon]$$

$$\leq \lim_{n \rightarrow \infty} \frac{1}{\varepsilon^2} E[|\bar{Y}_n - \mu|^2]$$

$$= \frac{1}{\varepsilon^2} \cdot \lim_{n \rightarrow \infty} \frac{1}{n} \cdot V[X] = 0$$

Recall:

$$E[|\bar{Y}_n - \mu|^2] = V[\bar{Y}_n]$$

$$= \frac{1}{n} V[Y_i]$$

$$= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(Y_i, Y_j)$$